Finding Signal in Network Simplicity

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***Abstract*—Engagement graphs, built from the retweets, replies and quotes that link accounts, are used to distinguish coordinated campaigns from organic activity. A model that classifies them well is easily read as having learned how campaigns spread, so the source of its predictive signal matters. How much of that signal can simple graph measurements capture, and what do structural and content features add? We study 103 Twitter event graphs from the Large Engagement Networks (LEN) project archive. We first compare node and edge counts with models adding degree concentration, higher-order arrangement and message content on complete graphs, and then ask what remains informative when only a fixed number of edges is observed, as in sampled engagement data. Node and edge counts alone separate campaign from comparison events well (balanced accuracy 0.884). The other features improved probabilistic predictions, lowering log loss in the prespecified comparison and in 19 of 20 repartitions, but classification decisions showed no detectable improvement, and no feature family alone detectably improved on the counts. Under a fixed edge budget, the number of distinct nodes a sample reaches remained highly informative, because it is largely set by the full graph's size and degree sequence. An edge-by-edge statistic, the dispersion of message token counts, measured the same quantity at every budget, yet its added predictive value, identified post hoc, was uncertain: measurement stability and additional predictive value are different questions. Complex features should therefore be judged against simple baselines, and stable measurements should not be read as additional predictive information.**

***Index Terms*—graph classification, engagement networks, coordinated campaigns, simple baselines, probabilistic evaluation, graph sampling**

I. Introduction

Coordinated campaigns on social media, such as organized efforts to push a hashtag into a platform's trending list, are often studied through engagement graphs, in which accounts are nodes and retweets, replies or quotes are directed edges. The Large Engagement Networks (LEN) benchmark [1] releases such graphs with event-level labels, and graph-learning methods have been proposed and evaluated on it [2]. A high score invites the reading that a model has learned something about how campaigns spread. That reading depends on what the labels can be predicted from, which makes it a question about what the benchmark measures and not only about how well models perform [3].

Graph-classification benchmarks are often matched by simple baselines that ignore much of the structure they are meant to test [4]–[6]. Engagement graphs give a specific reason to expect this. In the archive analysed here, the median comparison graph has 3,227 nodes and the median campaign graph 506. The first question is therefore what node and edge counts achieve on their own, and the second is what anything else adds to them.

Full graphs are also not the only way such data are seen. Engagement data are usually collected by sampling, and fixing the number of observed edges looks like a natural way to hold size constant across events. Sampling, however, changes graph statistics [7]. We use uniform edge sampling as a controlled probe of this second regime, not as a model of how any platform samples its data.

We ask how well the campaign label can be predicted from node and edge counts, and what, if anything, adds to that prediction. Under full observation (RQ1), we add degree concentration, higher-order arrangement and message content to the counts. Under edge-limited observation (RQ2), exactly K edges are seen, so the edge count is fixed by design and the count that still varies is the view's node count N_K; we ask what structural and message-level statistics add to it. The unit of analysis is the event graph, and every claim concerns predictive association. We make no claims about individual accounts, coordination mechanisms or learned graph models.

We study the 103-graph LEN project archive, obtained from the LEN project website [8]; Section III describes its provenance, and all results refer to this archive.

Three findings emerge. First, counts alone predict the label well. Second, adding all other features to the counts lowered log loss on the prespecified partition and in 19 of 20 repartitions, without a detectable gain in balanced accuracy; no single feature family detectably improved on the counts, and structural features helped only alongside content. Third, a fixed edge budget does not remove size. The view's node count is, in expectation, a function of the full graph's size and degree sequence and stays highly predictive, and the structural features of a view largely track it. A statistic computed edge by edge, token-count dispersion, keeps its meaning across budgets and was associated with lower log loss, but it was selected post hoc and its added value is uncertain. The paper contributes a count-first evaluation of nested feature sets with prespecified contrasts and repartition sensitivity; an edge-limited evaluation that identifies the count remaining in a fixed-budget view and why it remains; a comparison of collision-dependent and edge-by-edge statistics under sampling that keeps measurement agreement separate from predictive value; and a documented

description of the archive, including its text retention.

II. Related Work

LEN [1] releases engagement graphs labelled as campaign or non-campaign events, with campaign subtypes, and describes a smaller balanced subset, LEN-Small. Its authors note that comparison events tend to produce larger networks. DECODE [2] adds density-aware random-walk embeddings built on degree, core and truss numbers. We do not propose a detector or re-evaluate these models; we characterize the signal they are evaluated against.

Benchmark-effectiveness studies show that structure-agnostic or simple-property baselines can match graph neural networks [4], [5], and that dataset quality can be assessed by perturbing graph structure and features [6]. Degree-preserving rewiring is a standard null model for separating degree-sequence effects from higher-order arrangement [9], graph sampling is known to distort graph statistics [7], and small-sample cross-validation produces wide error bars [10]. We combine a count-first baseline, paired estimand-specific evaluation, a degree-preserving null and a fixed-budget observation design on the LEN project archive, and state what each analysis can and cannot show.

III. Data and Provenance

The archive was obtained from the official LEN project website [8] and contains 104 NetworkX node-link JSON files. Ingestion computes a SHA-256 digest of every file and removes 1 exact duplicate, a campaign graph whose two file names differ only in a Turkish dotless "ı" versus ASCII "i". The remaining 103 unique graphs comprise 52 campaign and 51 comparison events, labelled from the file-name tokens `_campaign_` and `_noncampaign_`. This archive is distinct from the dataset described in the published LEN paper, which reports LEN-Small as 100 graphs (51 campaign, 49 non-campaign) selected as the smallest by node count [1]; the two differ in membership and graph sizes. Throughout, we refer to the analysed collection as the 103-graph LEN project archive, and our results describe it rather than the published subset.

The graphs are directed and simple (no repeated directed pairs), with 288,540 edges in total, including 4,212 self-loops that occur in every event. Node identifiers are per-graph indices, so the release contains no account identifiers and cross-event account overlap cannot be measured. We exclude the pretrained text and profile embeddings attached to nodes and edges to keep the features interpretable. Each edge carries a message text, but 27.5% of the 288,540 texts are null. Campaign graphs retain less usable text than comparison graphs (mean usable share 0.58 against 0.80), 11 campaign graphs and zero comparison graphs keep under a fifth of their texts, and two campaign graphs have none. Content features therefore describe retained text only. Edge timestamps are present, but whether they record the engagement action or the engaged-with post is not verified, so no temporal features are used. Table I summarizes the archive, and Fig. 1 shows

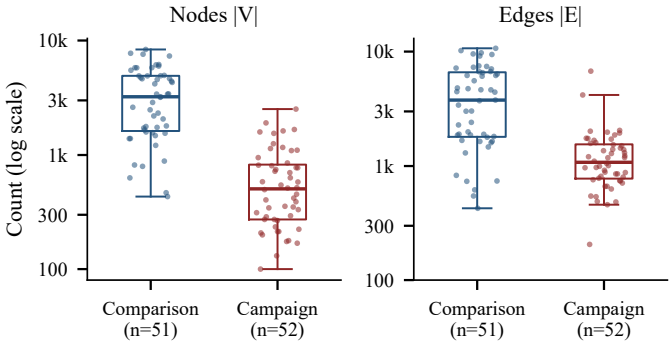


Fig. 1. Event-graph size by class on a logarithmic axis. Left: node count; right: edge count. Boxes show interquartile ranges and medians; points are events (comparison $n = 51$, campaign $n = 52$).

the size difference between classes (signed standardized mean difference of log node count -2.24).

TABLE I
Archive and Cohorts

Item	Value
Archive JSON files	104 (1 SHA-256 duplicate removed)
Canonical events	103 (52 campaign / 51 comparison)
LEN-Small (LEN paper)	100 graphs (51 / 49); a distinct dataset description (Section III)
Nodes, median [IQR]	campaign 506 [273–824]; comparison 3,227 [1,622–4,916]
Edges, median [IQR]	campaign 1,078 [776–1,548]; comparison 3,769 [1,799–6,595]
Edges / self-loops	288,540 / 4,212 (all 103 events)
Edge texts	288,540; 27.5% null
Usable-text share, mean	campaign 0.58; comparison 0.80; 2 campaign graphs have none
Edge-limited cohorts	$K = 250$: 102 (51/51); $K = 500$: 98 (48/50); $K = 1000$: 73 (28/45)
Account identifiers	none (per-graph node indices)

IV. Methods

A. Features

All features are computed per event graph $G = (V, E)$ with NetworkX [11] and listed in Table II. *Counts* are $\log(1 + |V|)$ and $\log(1 + |E|)$; density and mean degree are functions of them. *Degree concentration* is the Gini coefficient of the in-degree and of the out-degree sequence, where for a non-negative vector x sorted in ascending order

$$G(x) = \frac{2\sum_{i=1}^{n} i\, x_{(i)} - (n+1)\sum_{i=1}^{n} x_{(i)}}{n\sum_{i=1}^{n} x_{(i)}}. \qquad (1)$$

Arrangement comprises reciprocity, the fraction of nodes in the largest weakly connected component, and the median and maximum core number, computed on the simple undirected projection without self-loops. *Content* is computed from the edges' message texts. Texts are lower-cased, URLs (`https?://\S+`) are removed and whitespace is collapsed; null texts and texts empty after cleaning are dropped, and each remaining edge contributes its own text, so repeated messages are kept. Content comprises the exact-duplicate fraction (one

minus the share of unique texts), the Gini coefficient (1) of message frequencies, and *token-count dispersion*, the sample standard deviation of per-text whitespace token counts (0 for one text). A graph without usable text has undefined content features, which are imputed with the training-fold median. Recomputing all three content features from the raw archive, independently of the pipeline code, reproduced the stored values exactly.

The last column of Table II records how each feature behaves in a sample of edges. Most depend on how often sampled edges share endpoints, or sampled texts repeat, so their values in a view change with the budget. Token-count dispersion is an average over edges: a view estimates the same full-graph quantity at every budget.

TABLE II
FEATURE FAMILIES, MODELS AND BEHAVIOUR UNDER EDGE SAMPLING

Feature	**Models**	**In a K-edge view**
Log node count	all; A–D as $\log N_K$	shared endpoints
Log edge count	M0–M4	fixed at K
Squares and product of counts	M0q–M4	fixed or shared endpoints
In-/out-degree Gini	M1, M2, M4; C, D	shared endpoints
Reciprocity; largest weakly connected component; median, max core number	M2, M4; C, D	shared endpoints
Duplicate fraction; frequency Gini	M3, M4	repeated texts
Token-count dispersion	M3, M4; B, D	edge-by-edge average

B. Full Observation: Nested Models and Contrasts

Seven feature sets share identical event-level folds: M0, counts; M0q, counts with squared and interaction terms as a nonlinear count control; M1 = M0q + degree concentration; M2 = M1 + arrangement; M3 = M0q + content; M4 = M2 + content; and a diagnostic M4 + density. Three primary contrasts were specified before the final run: C1 (M4 − M0q, all features beyond counts), C2 (M2 − M1, arrangement beyond counts and degree concentration) and C3 (M3 − M0q, content beyond counts). They were not preregistered, and an earlier version of this study analysed the same archive.

The nesting is what makes each contrast interpretable. A larger model adds one family to a smaller one on the same events and folds, so a difference between them reflects the added family given the others, not a change of data or partition. The baseline for C1 and C3 is M0q rather than M0 because a feature that is a nonlinear function of size could otherwise appear to add information merely by capturing curvature in the counts; the squared and interaction terms themselves made no detectable difference (-0.005 [-0.016, +0.009]), so the choice of baseline does not drive the contrasts. Because content is added both to counts alone (M3) and to counts and structure (M4), the contribution of structure can be read in two ways: beyond counts alone (M2 − M0q) and beyond counts and content (M4 − M3). The two answers need not agree. In an additive model, disagreement shows that the families' contributions overlap or offset each other; it does not show an interaction.

Logistic regression (LR, with median imputation, standardization and balanced class weights) is the primary model [12]; gradient-boosted trees (HGBT) are secondary. Hyperparameters are tuned by inner three-fold cross-validation inside each of five stratified outer folds; for LR, the regularization strength C is chosen from $\{0.01, 0.1, 1, 10\}$. Log loss, area under the ROC curve (AUC) and balanced accuracy (BA) answer different questions, so BA comes from fits tuned for BA, and log loss and AUC from fits tuned for log loss; the regularization that maximizes BA at a fixed threshold need not give well-calibrated probabilities. We lead with log loss for every incremental comparison: it is a proper scoring rule, uses the whole predicted probability and does not depend on a threshold. AUC is reported as the ranking metric and BA as the decision metric. Both the BA and the log-loss tests of C1–C3 were prespecified, and both are reported; leading with log loss is a presentation choice.

C. Uncertainty, Repartitions and Status Labels

Differences are estimated on identical events with an event-level paired bootstrap of fixed out-of-fold predictions [13] (2,000 resamples, seed 7), and Holm correction is applied separately to the three BA and the three log-loss primary tests [14]. These intervals resample events for one partition of the events into folds (seed 42); they do not include the variability of refitting on a different partition [10]. As a sensitivity analysis, we repeated the full-observation analysis over 20 outer partitions (seeds 42 and 1–19), averaged each event's out-of-fold predictions over partitions, and bootstrapped those. The partitions reuse the same events and are not independent replications. Every result carries one of four labels: *primary* (C1–C3, and the models they compare, on the prespecified partition), *sensitivity* (the same question under different partitions, exclusions or fold grouping), *exploratory* (planned but outside the primary family) and *post hoc* (chosen after inspecting results on these events). Token-count dispersion was singled out after the full-observation results had been examined, so every analysis that isolates it is post hoc. Unless a Holm-adjusted p is given, intervals are unadjusted.

D. Degree-Preserving Null

To test arrangement beyond degree [9], each graph is rewired by directed double-edge swaps $(a, b), (c, d) \rightarrow (a, d), (c, b)$ on non-loop edges, rejecting swaps that create self-loops or repeated edges. This preserves every node's in- and out-degree, $|V|$, $|E|$ and the degree Gini coefficients exactly. We use 10 swaps per edge and 5 realizations per event, refit M2 with arrangement features from the rewired graphs, and compare it with M2 on the real graphs, with the rewired fits' probabilities averaged per event.

E. Edge-Limited Observation

For $K \in \{250, 500, 1000\}$, every event with at least K edges is reduced to K uniformly sampled edges, with

ten sampling seeds (100–109) per event, and every feature is recomputed from the sampled edges only. The cohorts therefore differ by budget (Table I); the 73 events eligible at $K = 1000$ form a common cohort eligible at every budget. $K = 2000$ is infeasible for five-fold evaluation because only 4 campaign events have at least 2,000 edges. All conditions at a given K use the same events, the full-cohort folds restricted to them, and the same feature definitions: *FF*, a model trained and evaluated on full graphs; *FK*, the full-graph model applied to single K-edge views (frozen transfer), each event being scored only by the fold model that held it out; and *KK*, a model trained and evaluated on single views from the same seed. The estimand is single-view performance, averaged over seeds, because a practitioner usually sees one sample of an event; an ensemble that averages probabilities over the ten views is reported separately, as a measure of what one sample's randomness costs. Contrasts use an event-level paired bootstrap (1,000 resamples). For the regime and feature contrasts, which answer separate questions, Holm-adjusted p-values are across the three budgets for each contrast and metric. The two token-count dispersion contrasts (B − A and D − C, defined below) test one post-hoc question at three budgets each, so their six p-values are adjusted together for each metric.

Three choices limit what this design can show. First, uniform sampling without replacement is the simplest observation process and makes the expected node count of a view computable exactly; platform collection is instead shaped by keywords, time windows and rate limits, and can over-represent active accounts or bursts of activity. The results describe a controlled probe, not any platform's sampling. Second, a view keeps only the nodes its sampled edges touch and the texts of those edges, so every view feature is computed from the view alone. Third, eligibility requires at least K edges, so larger budgets keep only larger events: comparisons across budgets mix observation depth with cohort composition, and the common cohort removes the composition change only for its 73 largest events.

In a view, $|E| = K$, and density $K/(N_K(N_K - 1))$ is a function of the view's node count N_K, the number of distinct endpoints of the sampled edges. Four models are compared within views: A, $\log N_K$ alone; B, A plus token-count dispersion; C, A plus the six structural features (degree Gini, reciprocity, largest component, coreness); and D, C plus token-count dispersion. B and D are post hoc. To explain N_K, we use its expectation under uniform sampling of K of the $m = |E|$ edges without replacement, where d_v is the number of distinct edges incident to v:

$$\mathbb{E}[N_K] = \sum_{v \in V} \left[1 - \binom{m - d_v}{K} \Big/ \binom{m}{K} \right]. \qquad (2)$$

This expectation depends only on K and the full graph's incidence-degree sequence.

F. Implementation and Verification

One pipeline runs every step from the raw archive to the results with fixed seeds. Every number in this paper is written into the manuscript by a script from the stored results, and the repository's tests fail if a result is typed by hand, if the generated values are stale, or if the paper and the project website disagree. An independent script reimplements the metrics, the bootstrap, Holm correction, the cross-validation and (2), and recomputes all pipeline comparisons from the saved predictions (21 of 21 checks pass). The repartition, exclusion, text-retention and token-count dispersion analyses were run by separate audit scripts that reuse the pipeline's stored features, folds and sampled edges; where the two overlap, the audit reproduces the pipeline's values exactly. Two implementation errors in an earlier internal version, coreness on graphs with self-loops and fold assignment for frozen models, were fixed before these results were produced and are covered by regression tests.

V. Results

A. Counts Alone

Node and edge counts alone predict the label well. LR on the two counts (M0) reaches BA 0.884, AUC 0.931 and log loss 0.331 on the prespecified partition (Table III), and node count alone reaches BA 0.874. Averaged over 20 partitions, the count model's BA is 0.864. Fig. 1 shows why counts predict well, and also why not perfectly: the classes overlap in the middle of the size range. If counts carry this much of the signal, the next question is what remains for every other feature.

TABLE III
Full Observation, LR: Out-of-Fold Performance by Feature Level ($N = 103$)

Level	Primary partition			20-partition mean		
	log loss	AUC	BA	log loss	AUC	BA
M0	0.331	0.931	0.884	0.328	0.932	0.864
M0q	0.326	0.932	0.884	0.325	0.933	0.867
M1	0.360	0.918	0.874	0.364	0.918	0.860
M2	0.352	0.932	0.864	0.326	0.935	0.873
M3	0.307	0.954	0.864	0.290	0.954	0.892
M4	0.216	0.971	0.893	0.231	0.967	0.902
M0q + disp.[P]	—	—	—	0.245	0.963	—

BA from BA-tuned fits; log loss and AUC from log-loss-tuned fits.
[P]Post hoc; not fitted on the primary partition alone.

B. What Adds to Counts Under Full Observation

All features together improve probabilities, not decisions. On the prespecified partition, adding degree, arrangement and content features to the count model (C1) changed log loss by −0.110 [−0.181, −0.039] (Holm-adjusted $p = 0.009$ across the three primary log-loss contrasts) and AUC by 0.039 [+0.009, +0.075] (unadjusted; Table IV). Because this interval does not include partition variability, we repeated the analysis over 20 outer partitions of the same events: the repeat-averaged difference was −0.094 [−0.172, −0.013] (unadjusted), and

TABLE IV
FULL OBSERVATION, LR: PAIRED CONTRASTS ON IDENTICAL EVENTS, Δ [95% INTERVAL]

Contrast	Primary partition (seed 42)			20 partitions, repeat-averaged		
	Δlog loss	**ΔAUC**	**ΔBA**	**Δlog loss**	**ΔAUC**	**Partitions favouring first model**
Primary: prespecified, Holm over the three tests for each metric						
C1: M4 − M0q all features beyond counts	$-0.110^{\dagger *}$ [−0.181, −0.039] (Holm 0.009)	$+0.039^{\dagger}$ [+0.009, +0.075]	+0.010 [−0.049, +0.076] (Holm 1.000)	$-0.094^{\dagger}$ [−0.172, −0.013]	$+0.034^{\dagger}$ [+0.006, +0.070]	19/20
C2: M2 − M1 arrangement beyond counts + degree	−0.008 [−0.079, +0.081] (Holm 1.000)	+0.014 [−0.019, +0.045]	−0.010 [−0.053, +0.032] (Holm 1.000)	−0.038 [−0.089, +0.016]	+0.017 [−0.008, +0.044]	20/20
C3: M3 − M0q content beyond counts	−0.019 [−0.136, +0.118] (Holm 1.000)	+0.022 [−0.012, +0.061]	−0.019 [−0.086, +0.046] (Holm 1.000)	−0.035 [−0.120, +0.069]	+0.022 [−0.007, +0.056]	17/20
Exploratory: conditional contributions, unadjusted						
M1 − M0q degree concentration beyond counts	$+0.034^{\dagger}$ [+0.006, +0.071]	$-0.014^{\dagger}$ [−0.028, −0.002]	−0.010 [−0.033, 0.000]	$+0.040^{\dagger}$ [+0.027, +0.053]	$-0.015^{\dagger}$ [−0.025, −0.007]	0/20
M2 − M0q structure beyond counts	—	—	—	+0.002 [−0.048, +0.053]	+0.002 [−0.021, +0.025]	9/20
M4 − M3 structure beyond counts + content	$-0.091^{\dagger}$ [−0.182, −0.018]	$+0.017^{\dagger}$ [+0.001, +0.037]	+0.029 [0.000, +0.067]	$-0.059^{\dagger}$ [−0.110, −0.011]	+0.013 [0.000, +0.027]	19/20
M4 − M2 content beyond counts + structure	$-0.136^{\dagger}$ [−0.235, −0.055]	$+0.039^{\dagger}$ [+0.009, +0.080]	+0.029 [−0.035, +0.091]	$-0.096^{\dagger}$ [−0.165, −0.032]	$+0.032^{\dagger}$ [+0.008, +0.064]	19/20
Sensitivity: C1 on altered event sets or folds, unadjusted						
C1 without the two no-text graphs ($n = 101$)	—	—	—	−0.083 [−0.171, +0.008]	$+0.033^{\dagger}$ [+0.004, +0.069]	18/20
C1 repeated hashtags in one fold	—	—	—	−0.082 [−0.164, +0.005]	$+0.030^{\dagger}$ [+0.002, +0.065]	19/20
Post hoc: token-count dispersion, unadjusted						
M0q + disp. − M0q dispersion beyond counts	—	—	—	−0.080 [−0.160, +0.004]	$+0.031^{\dagger}$ [+0.003, +0.068]	20/20
M4 − (M0q + disp.) all features beyond counts + disp.	—	—	—	−0.014 [−0.055, +0.024]	+0.004 [−0.009, +0.016]	16/20

Negative Δlog loss and positive ΔAUC or ΔBA favour the first model. Event-level paired bootstrap of fixed out-of-fold predictions (2,000 resamples). †95% interval excludes zero (nominal). *Holm-adjusted $p < 0.05$ within the named family. "—": not computed for that partitioning. The last column counts partitions in which the first model had lower log loss.

the direction was the same in 19 of the 20 partitions (range −0.122 to +0.028). The spread across partitions (standard deviation 0.035) is about as large as the event-level bootstrap uncertainty (0.040), so the single-partition interval omits a source of variation of comparable size. In BA, the primary C1 difference was +0.010 [−0.049, +0.076] (Holm $p = 1.000$): at the 0.5 threshold, M4 classified 6 events correctly that the count model missed, and missed 5 that it got right. The two results describe one pattern. The richer model moves most events' predicted probabilities toward the correct class, which log loss and AUC register, but few of those moves cross the decision threshold, so classifications barely change. The BA difference was positive in all 20 partitions (+0.010 to +0.059), but because the partitions reuse the same events this does not establish a reliable gain, and BA comparisons were sensitive to tuning tie-breaks: permuting the row order moved the C1 BA difference from +0.010 to +0.029, while the log-loss difference was unchanged.

No feature family improves on counts by itself, and structure helps only alongside content. Added to the count model alone, the six structural features produced no detectable change in log loss (M2 − M0q: +0.002 [−0.048, +0.053] over 20 partitions), and degree concentration on its own worsened log loss in every partition (+0.040 [+0.027, +0.053]). Arrangement beyond counts and degree concentration (C2) and the three content features beyond counts (C3) showed no detectable log-loss change on the prespecified partition (−0.008 [−0.079, +0.081] and −0.019 [−0.136, +0.118]; Holm $p = 1.000$ and 1.000). Added to a model that already contained counts and content, however, the same structural features changed log loss by −0.059 [−0.110, −0.011] (exploratory, unadjusted; 19 of 20 partitions; −0.091 [−0.182, −0.018] on the primary partition). Neither degree concentration nor arrangement alone accounts for this: added to counts, content and the other family, each gave an interval including zero ([−0.080, +0.006] and [−0.070, +0.057]). Content added to counts and structure (M4 − M2) changed log loss by −0.096 [−0.165, −0.032]. LR is additive, so these contrasts do not identify a modelled interaction, and they are predictive comparisons: they do not show that graph structure is uninformative, or informative in any causal sense.

The gain rests on a minority of events and on the treatment of missing text. Averaged over partitions, 82 of the 103 events had lower out-of-fold log loss under M4 than under the count model, but the ten events most favourable to M4 contributed 94% of the net gain. No single event drives it: refitting with each event left out kept the C1 log-loss difference between −0.106 and −0.070 (ten partitions). The two campaign graphs without usable text have imputed content features. Excluding them (101 events, 20 partitions), the C1 log-loss difference was −0.083 [−0.171, +0.008]: the same direction, but with an interval that includes zero. The AUC difference (+0.033

[+0.004, +0.069]) and the contrast for content beyond counts and structure (−0.087 [−0.158, −0.015]) kept intervals excluding zero (unadjusted). Keeping the two events from each repeated hashtag in one fold gave −0.082 [−0.164, +0.005].

The baseline and model choice matter for how firm the gain is. Widening the regularization grid made the count model's log loss worse (0.326 to 0.364), so under-tuning does not explain its disadvantage. Against more flexible count models, the gain is less firm: M4's log loss was lower than that of a natural-spline count model (−0.118 [−0.227, −0.011]) and a cubic count model (−0.090 [−0.165, −0.015]), but not detectably lower than quadratic discriminant analysis on the counts (−0.084 [−0.232, +0.081]), and none of these three comparisons had a Holm-adjusted p below 0.05 in the seven-contrast family assembled for this sensitivity analysis (0.132, 0.102 and 0.528). HGBT, the secondary model, gave a weaker count model (BA 0.805) and, for all features, a log loss (0.352) worse than LR's; against LR's count model, HGBT M4 changed BA by +0.048 [−0.018, +0.117].

A single feature, chosen post hoc, comes close to the full set. Token-count dispersion alone, added to the count model, changed log loss by −0.080 [−0.160, +0.004] and AUC by +0.031 [+0.003, +0.068], in the same direction in all 20 partitions. The full feature set did not detectably improve on counts plus this one feature (−0.014 [−0.055, +0.024]). Equivalence was not tested, and the feature was singled out after inspecting these results, so this does not show that dispersion accounts for the content gain.

Arrangement shows no detectable gain over its degree-preserving null. Real graphs differ strongly from their rewired counterparts (median z-score of reciprocity 5.96 in campaign and 26.11 in comparison events), but M2 with real arrangement features did not detectably outperform M2 with rewired ones: the differences were −0.037 [−0.123, +0.028] in log loss and 0.000 [−0.048, +0.047] in BA against the averaged rewired fits (real-graph BA 0.864; per-realization mean 0.876). The null is narrow. Rewiring changed reciprocity in 92% of events and the largest-component fraction in 99%, but the median core number in only 4%; about 13% of non-loop edges per graph could not be moved (median); and because it preserves the degree sequence, it preserves every quantity determined by it, including (2). In an exploratory analysis that matched campaign and comparison events one-to-one on log node count (15–17 pairs over visiting orders, 16 in the primary one), the count model fell to near chance (median BA over visiting orders 0.531) while the content model's median was 0.781; with so few pairs these are point estimates without sampling intervals, and matching changes the evaluated population.

If counts carry most of the signal under full observation, does fixing the number of observed edges remove it?

C. Edge-Limited Observation: The Count That Remains

The view's node count stays highly predictive. In a K-edge view the edge count is fixed, but N_K alone (model A, KK) reaches AUC 0.961, 0.952 and 0.890 at $K = 250$, 500 and 1000 (Table V). The expectation (2) matches the observed mean N_K within 3.5% at $K = 250$ and agrees with a Monte Carlo check on three events. Computed directly from the score, without fitting a model, its AUC is 0.965, against 0.934 for full-graph log node count. Full-graph node count and mean degree explain 93.0% of the variation of $\log \mathbb{E}[N_{250}]$ (node count alone 77.2%), and in/out-degree Gini adds nothing measurable (93.0%; Fig. 2). A fixed edge budget therefore does not hold size constant: the number of distinct nodes a view reaches re-encodes the full graph's size and degree sequence.

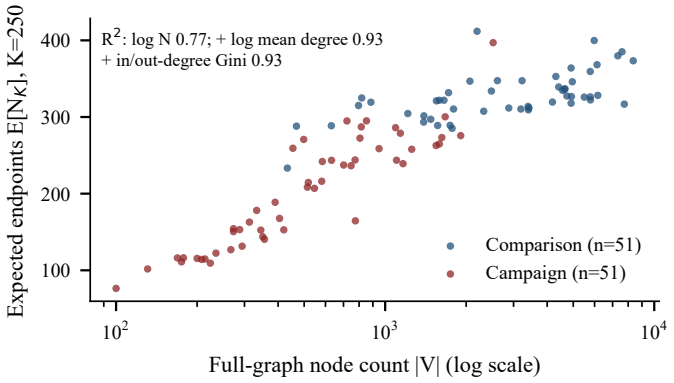


Fig. 2. Expected distinct endpoints in a 250-edge sample, from (2), against full-graph node count for the 102 eligible events. Inset: variance of $\log \mathbb{E}[N_K]$ explained by log node count, by log node count and log mean degree, and additionally by in/out-degree Gini.

Structural features in a view largely track N_K. Within views, a linear regression of $\log N_K$ on the six structural features, cross-validated with events grouped, reaches R^2 of 0.921, 0.954 and 0.937 at the three budgets. Adding them to N_K (C − A) produced no detectable change in log loss at any budget (−0.006 [−0.079, +0.062], +0.013 [−0.049, +0.088] and +0.041 [−0.033, +0.129]; Table VI). The three content features, added to N_K and structure, changed log loss by −0.067, −0.127 and −0.173, with intervals excluding zero at $K = 500$ and 1000 (Holm-adjusted p <0.003 and <0.003), while their BA intervals included zero at every budget.

A single view was not detectably worse than the full graph. With all features, a model trained and evaluated on single views (KK) and the full-graph model on the same events (FF) differed in log loss by −0.022 [−0.072, +0.035], −0.032 [−0.074, +0.005] and −0.005 [−0.060, +0.046]. In BA the view-trained model was ahead at $K = 1000$ (+0.056 [+0.009, +0.114], nominal), but its AUC was almost the same (0.953 against 0.951), so that difference reflects where the decision threshold falls rather than better ranking; it comes from a small, imbalanced cohort (73 events, 28/45) and does not suggest that sampling improves classification.

Retraining on views matters because view features measure something different. Applying the full-graph model to views (FK) degrades predictions, and the reason is visible in the metrics. For N_K alone, frozen transfer leaves ranking largely intact (AUC 0.942 at $K = 250$) but collapses decisions

to a single class (BA 0.500), because a view's node count sits on a different scale from a full graph's. For the six structural features, frozen transfer also loses ranking (AUC 0.693, against 0.947 when trained on views), and lost 0.126 in BA relative to full graphs. Training on views instead changed log loss by −0.338, −0.205 and −0.148 at the three budgets (each Holm-adjusted $p < 0.05$) and BA by 0.254 [0.162, 0.348] at $K = 250$. Because these features largely track N_K within views, the contrast mixes the training regime with a change in what the features measure. Averaging the ten views changed BA by at most 0.004 and lowered log loss by at most 0.022: detectable, but small. On the common cohort of 73 events, the retraining, transfer, content and ensembling contrasts have the same log-loss directions at $K = 250$ and 500. The structure contrast does not: at $K = 250$ it changes sign, from −0.006 [−0.079, +0.062] on the budget-specific cohort to +0.049 [−0.015, +0.137] on the common cohort. Both intervals include zero, so this is a change in an estimate that was indistinguishable from zero on either cohort, not a reversal of a detected effect.

Is there a statistic whose meaning does not change with the budget, and does it add anything to N_K?

D. Token-Count Dispersion Under Edge Limits

Measurement. Because token-count dispersion is an average over edges, its view values tracked its full-graph values closely (Spearman $\rho \geq 0.995$ between the seed-averaged view value and the full-graph value at every budget). This is the expected behaviour of such a statistic estimated from hundreds of edges, and at $K = 1000$ a view covers most edges of a typical campaign graph, so it is a measurement check rather than a finding. Comparison events had higher dispersion: its univariate AUC was 0.78, 0.77 and 0.81 in views at the three budgets, the same as on the full graphs of those events (0.78, 0.77, 0.81).

Incremental value (post hoc). The question here is whether a view's message lengths add anything once its node count, or its node count and structure, are known. Adding dispersion to N_K (B − A) changed single-view log loss by −0.036 [−0.112, +0.038], −0.068 [−0.163, +0.018] and −0.114 [−0.218, −0.011] at $K = 250$, 500 and 1000 (Fig. 3). In every budget and contrast, at least 10 of the 10 sampling seeds favoured the model with dispersion, but the seeds share events, and the B − A intervals include zero at $K = 250$ and 500. At $K = 1000$ (73 events), dispersion was associated with lower log loss both when added to N_K (B − A, nominal interval excluding zero, Holm-adjusted $p = 0.140$) and when added to N_K and structure (D − C: −0.138 [−0.208, −0.076], Holm-adjusted p <0.006), with AUC differences of +0.065 [+0.010, +0.144] and +0.045 [+0.002, +0.111]. After Holm adjustment over the six dispersion contrasts, only D − C at $K = 1000$ keeps a log-loss difference with adjusted $p < 0.05$; the other log-loss differences are nominal at best. In BA, B − A at $K = 500$ also survives adjustment (Table VI), but its AUC difference is +0.002 [−0.037, +0.038], so that gain reflects where the threshold falls rather than better ranking. We did

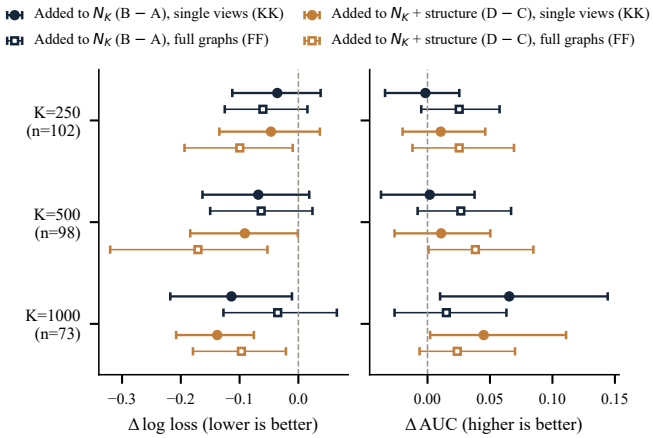


Fig. 3. Post hoc: change in single-view log loss (left) and AUC (right) when token-count dispersion is added to N_K (B − A) or to N_K and structure (D − C), by edge budget. Filled circles: trained and evaluated on single views (KK); open squares: the same contrasts on full graphs of the same events (FF). Bars are 95% event-level paired bootstrap intervals, unadjusted. Cohort sizes differ by budget.

not test whether the two contrasts differ, and their intervals overlap almost entirely. Model C, with N_K and structure but without dispersion, had the highest log loss of the four at $K = 1000$ (0.423, against 0.382 for N_K alone), so D − C is measured against a comparatively poorly calibrated baseline. D − C was −0.047 [−0.134, +0.037] at $K = 250$ and −0.091 [−0.184, −0.001] at $K = 500$, the latter interval only narrowly excluding zero. Cohort composition changes with K, and B and D were not computed on the common cohort below $K = 1000$, so these results do not show that the added value of dispersion depends on the budget.

Text availability. Retained text differs by class, and dispersion is not independent of it. Within comparison events, views with more usable text had lower dispersion (event-level Spearman ρ from −0.356 to −0.504 across budgets). Within strata of the usable-text share, dispersion still separated the classes (AUC 0.77–0.93; descriptive). Excluding the two graphs without usable text changed the six B − A and D − C log-loss differences by at most 0.014 in five of them and by 0.028 in the remaining one.

VI. Discussion

Table VII summarizes the main findings with their status. The firm results concern counts; the conditional and edge-limited results are exploratory; everything about token-count dispersion is post hoc.

Counts first. In this archive, most of the predictability comes from node and edge counts. Why size should carry so much of the label is not resolved by these data, and the possible explanations have different implications. Organic trends can draw large and diverse audiences, while a campaign hashtag pushed by a limited group over a short period may reach fewer accounts; the LEN authors note that comparison events tend to produce larger networks [1]. It could also reflect how the events in this archive were selected. Under the first

TABLE V
EDGE-LIMITED OBSERVATION: SINGLE-VIEW PERFORMANCE BY CONDITION AND FEATURE SET (BUDGET-SPECIFIC COHORTS)

	$K = 250$ ($n = 102$; **51/51**)			$K = 500$ ($n = 98$; **48/50**)			$K = 1000$ ($n = 73$; **28/45**)		
Condition, features	**log loss**	**AUC**	**BA**	**log loss**	**AUC**	**BA**	**log loss**	**AUC**	**BA**
FF, all features	0.228	0.965	0.931	0.232	0.965	0.907	0.255	0.951	0.860
FF, node count	0.331	0.931	0.873	0.337	0.930	0.887	0.262	0.952	0.855
FF, structure	0.510	0.767	0.775	0.515	0.742	0.772	0.593	0.656	0.739
FK, node count	1.584	0.942	0.500	1.119	0.885	0.500	1.123	0.905	0.482
FK, structure	0.616	0.693	0.648	0.536	0.756	0.743	0.565	0.718	0.728
KK, structure	0.278	0.947	0.902	0.331	0.939	0.895	0.418	0.891	0.873
KK, N_K (A)	0.279	0.961	0.922	0.315	0.952	0.880	0.382	0.890	0.851
KK, N_K + structure (C)	0.273	0.946	0.914	0.328	0.944	0.905	0.423	0.899	0.885
KK, N_K + structure + content	0.206	0.967	0.933	0.200	0.969	0.934	0.250	0.953	0.915
KK, N_K + dispersion (B)[P]	0.243	0.959	0.950	0.246	0.954	0.949	0.268	0.955	0.942
KK, N_K + structure + disp. (D)[P]	0.226	0.957	0.933	0.237	0.955	0.942	0.285	0.944	0.917
KK, all features, 10-view ensemble	0.184	0.975	0.931	0.194	0.970	0.938	0.247	0.953	0.917

Means over ten sampling seeds. FF: trained and evaluated on full graphs of the same events (no sampling). FK: full-graph model applied to single views. KK: trained and evaluated on single views. Node count is $\log |V|$ for FF and $\log N_K$ in views; structure is the six structural features. [P]Post hoc.

TABLE VI
EDGE-LIMITED OBSERVATION: PAIRED CONTRASTS, Δ [95% INTERVAL], LOG LOSS FIRST

	$K = 250$		$K = 500$		$K = 1000$	
Contrast	**Δlog loss**	**ΔBA**	**Δlog loss**	**ΔBA**	**Δlog loss**	**ΔBA**
Observation regime (exploratory)						
View − full graph (KK − FF), all features	−0.022 [−0.072, +0.035]	+0.002 [−0.031, +0.038]	−0.032 [−0.074, +0.005]	+0.027 [−0.003, +0.065]	−0.005 [−0.060, +0.046]	+0.056† [+0.009, +0.114]
Retraining (KK − FK), structure	−0.338$^{\dagger *}$ [−0.464, −0.213]	+0.254$^{\dagger *}$ [+0.162, +0.348]	−0.205$^{\dagger *}$ [−0.341, −0.059]	+0.152$^{\dagger *}$ [+0.084, +0.226]	−0.148$^{\dagger *}$ [−0.259, −0.016]	+0.145$^{\dagger *}$ [+0.058, +0.236]
Retraining (KK − FK), all features	−0.562$^{\dagger *}$ [−0.834, −0.329]	+0.302$^{\dagger *}$ [+0.235, +0.368]	−0.180$^{\dagger *}$ [−0.276, −0.079]	+0.155$^{\dagger *}$ [+0.088, +0.228]	−0.144$^{\dagger *}$ [−0.236, −0.045]	+0.062 [−0.009, +0.133]
Transfer (FK − FF), structure	+0.106$^{\dagger *}$ [+0.056, +0.165]	−0.126$^{\dagger *}$ [−0.197, −0.068]	+0.021 [−0.016, +0.057]	−0.029 [−0.071, +0.009]	−0.027 [−0.061, +0.003]	−0.010 [−0.029, 0.000]
10-view ensemble − single view, all	−0.022$^{\dagger *}$ [−0.044, −0.007]	−0.002 [−0.019, +0.011]	−0.006$^{\dagger *}$ [−0.011, −0.002]	+0.004 [0.000, +0.009]	−0.002$^{\dagger *}$ [−0.004, −0.001]	+0.002 [−0.012, +0.016]
Features within views, KK (exploratory)						
Structure beyond N_K (C − A)	−0.006 [−0.079, +0.062]	−0.008 [−0.033, +0.019]	+0.013 [−0.049, +0.088]	+0.026 [−0.013, +0.076]	+0.041 [−0.033, +0.129]	+0.034 [−0.046, +0.113]
Content beyond N_K + structure	−0.067 [−0.145, +0.005]	+0.020 [−0.010, +0.049]	−0.127$^{\dagger *}$ [−0.226, −0.036]	+0.029 [−0.003, +0.062]	−0.173$^{\dagger *}$ [−0.247, −0.105]	+0.030 [−0.013, +0.084]
Token-count dispersion within views, KK (post hoc)						
Dispersion beyond N_K (B − A)	−0.036 [−0.112, +0.038]	+0.028 [−0.007, +0.066]	−0.068 [−0.163, +0.018]	+0.070$^{\dagger *}$ [+0.020, +0.127]	−0.114† [−0.218, −0.011]	+0.091† [+0.010, +0.176]
Dispersion beyond N_K + structure (D − C)	−0.047 [−0.134, +0.037]	+0.020 [−0.008, +0.047]	−0.091† [−0.184, −0.001]	+0.037† [+0.003, +0.075]	−0.138$^{\dagger *}$ [−0.208, −0.076]	+0.031 [−0.010, +0.086]

Budget-specific cohorts (Table V); single-view estimand. Event-level paired bootstrap (1,000 resamples). †95% interval excludes zero (nominal). *Holm-adjusted $p < 0.05$, across the three budgets for each regime or feature contrast and metric, and across all six dispersion contrasts for each metric; the view-minus-full-graph contrast has no stored p-value and carries only the nominal marker. Interval-based ΔAUC was computed only for the token-count dispersion contrasts (Fig. 3); condition-level AUC is in Table V.

explanation, size is a genuine property of the events that any model will exploit; under the second, it is an artefact of the data that would not transfer. In either case, a model that learns event size learns something real about these labels without necessarily learning anything about coordination.

The other features improved probabilistic prediction, not decisions, and the improvement is concentrated in a minority of events and sensitive to how missing text and flexible count models are handled. Evaluations on these data should therefore report a count-only baseline and report probability and decision metrics separately: a richer model can sharpen probabilities without changing many classifications.

Structure is conditional, not absent. Structural features showed no detectable change when added to counts or to N_K, but improved log loss alongside content. The degree-preserving null tests a narrow question, because rewiring preserves everything the degree sequence determines, including the expected node count of a view. None of these results shows that graph structure is uninformative; they show that, for the features and model used, its contribution beyond counts was not detectable on its own.

What remains predictive under a fixed edge budget. Fixing the number of observed edges looks like a way to compare events of different sizes on equal terms, but it does not remove size. A view's node count is, in expectation, a function of the budget and the full graph's degree sequence: a large graph whose edges spread over many accounts yields many distinct endpoints in a sample, and a small, concentrated graph yields

TABLE VII
MAIN FINDINGS AND THE STRENGTH OF THEIR EVIDENCE

Finding	Status	Evidence
Counts predict the label well	Primary	Count models M0 and M0q: all three metrics, every partition
All features improve log loss and AUC over counts	Primary; sensitivity	Holm-adjusted on the prespecified partition; 19 of 20 repartitions; interval includes zero without the no-text graphs
No detectable BA gain from added features	Primary	Inconclusive, not evidence of equivalence
Structure adds only alongside content	Exploratory	Unadjusted; not attributable to one family
N_K stays predictive in fixed-K views	Exploratory	Every budget; expectation exact under uniform sampling
Structure adds nothing detectable to N_K	Exploratory	Intervals include zero at every budget
Dispersion adds to N_K	Post hoc	Consistent direction; adjusted support only for D − C at $K = 1000$

few. Size therefore re-enters through a quantity that varies even though the edge count is held fixed. The structural features computed on a view largely track that count, which explains why views remain predictive, why structure adds nothing detectable to N_K, and why full-graph models transfer poorly to views. Anyone who uses fixed-budget samples to control for size should check what the samples' node counts alone achieve. Uniform edge sampling is a probe, and platform sampling may behave differently.

Measurement stability is not predictive value. These results separate two properties that are easy to conflate. A statistic is stable under sampling if its value in a view estimates its full-graph value: token-count dispersion is, and the structural features and N_K are not. Stability decides whether a model trained on full graphs can be applied to samples, and its absence explains the failure of frozen transfer. It says nothing about whether a feature adds information beyond what a view already offers. N_K is unstable across budgets yet highly predictive within each, while dispersion is stable but its increment over N_K is uncertain.

What token-count dispersion is and is not. Dispersion is a statistic of how much message lengths, in whitespace tokens, vary within the retained text of an event. Whether it adds predictive value beyond N_K is unclear: the direction was consistent, adjusted support is limited to one contrast at the largest budget, it rests on retained text, and it was selected after inspecting these events. Topic or event-type differences, text retention and cleaning choices are alternative explanations. It is not evidence of coordination or templated messaging. Testing it requires independent events and a common-cohort evaluation across budgets.

VII. LIMITATIONS

Scope and labels. All results describe the 103-graph LEN project archive (Section III); size–label associations may reflect how its events were selected and need not hold in other LEN data. Labels come from file names and were not independently verified; the comparison class means "not labelled a campaign", not verified organic activity. Campaign subtypes are not available, so heterogeneity across campaign types cannot be examined.

Sample size and dependence. With 103 events (73 at $K = 1000$), intervals are wide, and a result showing no detectable difference is not evidence of equivalence. Bootstrap intervals exclude refitting variability, which the repartitions show to be of comparable size. Two campaign hashtags each appear as two events on nearby dates, and one pair falls in different folds of the primary partition. Without account identifiers, dependence between events through shared accounts cannot be measured.

Selection. Token-count dispersion was chosen on these events, and no independent sample was available to test it. Holm adjustment controls for testing it at several budgets and baselines, not for having chosen it after looking at the data.

Missing text. All content features are computed on retained text. Retention differs by class, two campaign graphs have none and receive imputed values, and why texts are missing is unknown: deleted posts, suspended accounts and collection choices are all possible, and if any of them affects campaigns more than organic events, missingness would itself carry label information. Within comparison events, dispersion is associated with how much text a view retains, so what dispersion measures cannot be fully separated from retention.

Cohorts and sampled views. In the edge-limited analysis, budget is confounded with cohort composition: larger budgets keep only larger events, and the 73-event cohort at $K = 1000$ is imbalanced. B and D were not computed on the common cohort below $K = 1000$, so no budget trend can be claimed for dispersion. Only uniform edge sampling was studied, and a view's features use only its sampled edges, which is a stricter and simpler observation model than platform collection.

Models and null. The degree-preserving null mixes incompletely in hub-dominated graphs, rarely changes the median core number and uses 5 realizations. Only LR and HGBT on hand-crafted features were evaluated; graph neural networks and the methods of [1], [2] were not re-run, and published LEN-Small accuracies use a different protocol and metric. Timestamp semantics are unverified, so no temporal features are used.

VIII. CONCLUSION

In this archive of 103 engagement graphs, node and edge counts predict the campaign label well. Adding degree, arrangement and content features improved probabilistic prediction on the prespecified partition and in most repartitions, but showed no detectable gain in classification decisions, and no single family detectably improved on the counts by itself. Under a fixed edge budget, the view's node count re-encodes the full graph's size and degree sequence and remains highly predictive, while an edge-by-edge statistic, token-count dispersion, keeps its meaning across budgets and is a lead,

chosen post hoc, for independent testing. Two next steps follow: repeating the analysis on the published LEN-Small selection and other LEN data, and evaluating token-count dispersion and learned graph models on independent events under the same paired protocol.

Code and Data Availability

Code, configuration, tests, derived outputs, the audit record (`research_audit`) and reproduction instructions are at https://github.com/SynshAmi/temporal-coordination-in-social-media; the MIT licence covers the code only, not this paper, its figures or any data. A complete rerun of the pipeline reproduced every pipeline value on the original machine; other hardware has not been tested, and the audit scripts were run once. LEN is released by its authors under a CC-BY licence [1]. We do not redistribute the raw archive, which must be obtained from the LEN project [8]; the repository records its SHA-256 digest. Derived files (event names, counts, file digests, features and predictions) and, for the project website, each event's graph structure (node indices and directed edges, without text or timestamps) are shared with attribution and are not endorsed by the LEN authors.

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